

AQUABC Model

Parameter & Constant Reference

Based on WCONST_04.txt (latest configuration)

Total constants: 323

Generated: February 2026

This document provides a complete reference for all model constants used in the AQUABC pelagic ecological model (file WCONST_04.txt). Constants are organised by functional category. Each section includes the constant number, name, current value, description, and relevant literature references that justify the chosen parameterisation. The AQUABC model is a fully featured water quality / eutrophication model that simulates phytoplankton dynamics (diatoms, cyanobacteria, other algae, Nostocales), zooplankton grazing, organic matter cycling, nutrient dynamics (N, P, Si), dissolved oxygen, redox chemistry (Fe, Mn), methane, hydrogen sulphide, and pH/alkalinity. Temperature effects use the Cardinal Temperature Model with Inflection (CTMI; Rosso et al. 1993). Nutrient limitation follows Monod (Michaelis-Menten) kinetics. The multi-electron-acceptor mineralisation scheme follows the thermodynamic sequence of Van Cappellen & Wang (1996).

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General / Physical Parameters

General physical and optical parameters controlling aeration and light.

Typical units: mg O₂/L/day, 1/m/(ug Chla/L), mg C/mole photon

#	Constant Name	Value	Description
1	1	K_A	Aeration coefficient (if negative calculates internally)
2	2	THETA_K_A	Temperature correction factor for aeration
3	3	XKC	fixme. Light extinction per chlorophyl unit,(mcg Chla/l/m) for calculation of saturation for Smith. Should be different for different phyto?
4	4	PHIMX	Quantum yield const. mg C/mole photon

Literature References:

[1] Chapra, S.C. (1997)

Surface Water-Quality Modeling. McGraw-Hill, New York.

Relevance: Aeration coefficients and temperature correction theta = 1.024 for reaeration.

[2] Bowie, G.L., Mills, W.B., Porcella, D.B. et al. (1985)

Rates, Constants, and Kinetics Formulations in Surface Water Quality Modeling (2nd ed.). EPA/600/3-85/040. U.S. EPA, Athens, GA.

Relevance: Comprehensive compilation of rate constants for surface water quality models.

[3] Kirk, J.T.O. (1994)

Light and Photosynthesis in Aquatic Ecosystems, 2nd ed. Cambridge Univ. Press.

Relevance: Light extinction coefficient per unit chlorophyll (XKC ~ 0.01-0.02 m⁻¹/(ug Chla/L)).

[4] Falkowski, P.G. & Raven, J.A. (2007)

Aquatic Photosynthesis, 2nd ed. Princeton Univ. Press.

Relevance: Quantum yield constant PHIMX (700-1000 mg C/mole photon).

Diatoms (Bacillariophyceae)

Growth, respiration, mortality, nutrient limitation, and stoichiometry for diatoms -- siliceous phytoplankton that dominate spring blooms.
Typical units: 1/day, deg C, mg/L, dimensionless

#	Constant Name	Value	Description
5	5	KG_DIA_OPT_TEMP	3.1 Diatoms Growth rate
6	6	DIA_OPT_TEMP	Diatoms CTMI T_min (minimum cardinal temperature)
7	7	DIA_OPT_TEMP	Diatoms CTMI T_opt (optimal temperature)
8	8	EFF_DIA_GROWTH	Diatoms Effective growth. (1-EG)*growth - losses for respiration and excretion
9	9	KAPPA_DIA_UNDER_OP	(unused, kept for backward compat)
10	10	KAPPA_DIA_OVER_OP	Diatoms CTMI T_max (maximum cardinal temperature)
11	11	KR_DIA_20	Diatoms Respiration rate
12	12	THETA_KR_DIA	Diatoms Temperature correction for basal respiration rate
13	13	KD_DIA_20	Diatoms Mortality rate
14	14	THETA_KD_DIA	Diatoms Temperature correction for Mortality rate
15	15	KHS_DIN_DIA	Diatoms Half saturation growth for DIN
16	16	KHS_DIP_DIA	Diatoms Half saturation growth for DIP
17	17	KHS_DSi_DIA	Diatoms Half saturation growth for DSi
18	18	KHS_O2_DIA	Diatoms Half saturation growth for O2
19	19	FRAC_DIA_EXC	Diatoms Fraction of excretion in metabolism rate
20	20	I_S_DIA	Diatoms Light saturation (langleys)
21	21	DO_STR_HYPOX_DIA_D	Diatoms Dissolved oxygen stress in oxygen units (mortality increase below this value exponentially)
22	22	THETA_HYPOX_DIA	Diatoms Multiplier of the exponent for Dissolved oxygen stress
23	23	EXPON_HYPOX_DIA	Diatoms Exponent constant for Dissolved oxygen stress
24	24	DIA_N_TO_C	Diatoms Nitrogen to Carbon ratio
25	25	DIA_P_TO_C	Diatoms Phosphorus to Carbon ratio
26	26	DIA_Si_TO_C	Diatoms Silica to Carbon ratio
27	27	DIA_O2_TO_C	Diatoms Oxygen to Carbon ratio for respiration
28	28	DIA_C_TO_CHL	Diatoms Carbon to Chlorophyll a ratio

Literature References:

[1] Eppley, R.W. (1972)

Temperature and phytoplankton growth in the sea. Fishery Bulletin, 70(4), 1063-1085.

Relevance: Maximum growth rate as a function of temperature; diatom μ_{max} ~ 2-4 /day at optimal T.

[2] Rosso, L., Lobry, J.R., & Flandrois, J.P. (1993)

An unexpected correlation between cardinal temperatures of microbial growth highlighted by a new model. J. Theoretical Biology, 162(4), 447-463.

Relevance: Cardinal Temperature Model with Inflection (CTMI): T_min, T_opt, T_max framework used for all phytoplankton temperature responses in AQUABC.

[3] Bernard, O. & Remond, B. (2012)

Validation of a simple model accounting for light and temperature effect on microalgal growth. Bioresource Technology, 123, 520-527.

Relevance: Validated CTMI with three cardinal temperatures for various microalgae.

[4] Brzezinski, M.A. (1985)

The Si:C:N ratio of marine diatoms: interspecific variability and the effect of some environmental variables. J. Phycology, 21(3), 347-357.

Relevance: Si:C ratio for diatoms ~ 0.13-0.47 (model uses 0.25); Si:N:P = 15:16:1.

[5] Redfield, A.C. (1958)

The biological control of chemical factors in the environment. American Scientist, 46(3), 205-221.

Relevance: Redfield ratio C:N:P = 106:16:1 (N:C ~ 0.17, P:C ~ 0.024 by mass with molar conversion). Model N:C = 0.22 and P:C = 0.024 are within reported ranges.

[6] Geider, R.J., MacIntyre, H.L., & Kana, T.M. (1997)

Dynamic model of phytoplankton growth and acclimation: responses of the balanced growth rate and the chlorophyll a:carbon ratio. Marine Ecology Progress Series, 148, 187-200.

Relevance: C:Chl-a ratio 20-100 mg C / mg Chla (model uses 30 for diatoms).

[7] Reynolds, C.S. (2006)

The Ecology of Phytoplankton. Cambridge Univ. Press.

Relevance: Diatom respiration 0.03-0.10 /day; mortality 0.05-0.20 /day; KHS_DIN 0.005-0.05 mg/L.

Non-Fixing Cyanobacteria

Parameters for non-nitrogen-fixing cyanobacteria (e.g. *Microcystis*, *Planktothrix*). Warm-water specialists with higher optimal temperatures.

Typical units: 1/day, deg C, mg/L, dimensionless

#	Constant Name	Value	Description
29	29	KG_CYN_OPT_TE	Non-fixing cyanobacteria Growth rate
30	30	CYN_OPT_TEMP	Non-fixing cyanobacteria CTMI T_min
31	31	CYN_OPT_TEMP	Non-fixing cyanobacteria CTMI T_opt
32	32	EFF_CYN_GROWTH	Non-fixing cyanobacteria Effective growth. (1-EG)*growth - losses for respiration and excretion
33	33	KAPPA_CYN_UNDER_OP	(unused)
34	34	KAPPA_CYN_OVER_OP	Non-fixing cyanobacteria CTMI T_max
35	35	KR_CYN_20	Non-fixing cyanobacteria Respiration rate
36	36	THETA_KR_CYN	Non-fixing cyanobacteria Temperature correction for respiration rate
37	37	KD_CYN_20	Non-fixing cyanobacteria Mortality rate
38	38	THETA_KD_CYN	Non-fixing cyanobacteria Temperature correction for Mortality rate
39	39	KHS_DIN_CYN	Non-fixing cyanobacteria Half saturation growth for DIN
40	40	KHS_DIP_CYN	Non-fixing cyanobacteria Half saturation growth for DIP
41	41	KHS_O2_CYN	Non-fixing cyanobacteria Half saturation growth for O2
42	42	FRAC_CYN_EXC	Non-fixing cyanobacteria Fraction of excretion in metabolism rate
43	43	I_S_CYN	Non-fixing cyanobacteria Light saturation (langleys).Not used, is calculated!
44	44	DO_STR_HYPOX_CYN	Non-fixing cyanobacteria Dissolved oxygen stress in oxygen units (mortality increase below this value exponentially)
45	45	THETA_HYPOX_CYN	Non-fixing cyanobacteria Multiplier of the exponent for Dissolved oxygen stress
46	46	EXPON_HYPOX_CYN	Non-fixing cyanobacteria Exponent constant for Dissolved oxygen stress
47	47	CYN_N_TO_C	Non-fixing cyanobacteria Nitrogen to Carbon ratio ,was 0.1
48	48	CYN_P_TO_C	Non-fixing cyanobacteria Phosphorus to Carbon ratio
49	49	CYN_O2_TO_C	Non-fixing cyanobacteria Oxygen to Carbon ratio for respiration
50	50	CYN_C_TO_CHL	Non-fixing cyanobacteria Carbon to Chlorophyl a ratio

Literature References:

[1] Robarts, R.D. & Zohary, T. (1987)

Temperature effects on photosynthetic capacity, respiration, and growth rates of bloom-forming cyanobacteria. New Zealand J. Marine and Freshwater Research, 21(3), 391-399.

Relevance: Optimal growth temperature for *Microcystis* 25-30 deg C.

[2] Reynolds, C.S. (2006)

The Ecology of Phytoplankton. Cambridge Univ. Press.

Relevance: Cyanobacteria max growth rates 1.0-3.5 /day; C:Chl-a = 30-60.

[3] Rosso, L. et al. (1993)

As above -- CTMI model for temperature response.

Relevance: T_min = 10-18 deg C, T_opt = 25-30 deg C, T_max = 35-42 deg C for cyanobacteria.

[4] Jorgensen, S.E. et al. (1991)

Handbook of Ecological Parameters and Ecotoxicology. Elsevier.

Relevance: Half-saturation constants for DIN (0.005-0.05 mg/L) and DIP (0.003-0.015 mg/L).

Nitrogen-Fixing Cyanobacteria

Parameters for heterocystous N₂-fixing cyanobacteria (e.g. *Aphanizomenon*, *Dolichospermum*). Unique N-fixation parameters R_{FIX} and K_{FIX} .

Typical units: 1/day, deg C, mg/L, dimensionless

#	Constant Name	Value	Description
51	51	KG_FIX_CYN_OPT_T	Fixing cyanobacteria Growth rate constant
52	52	FIX_CYN_OPT_TEM	Fixing Cyanobacteria CTMI T _{min}
53	53	FIX_CYN_OPT_TEM	Fixing Cyanobacteria CTMI T _{opt}
54	54	EFF_FIX_CYN_GROWTH	Fixing cyanobacteria Effective growth. (1-EG)*growth - losses for RESP and excretion
55	55	KAPPA_FIX_CYN_UNDER	(unused)
56	56	KAPPA_FIX_CYN_OVER	Fixing cyanobacteria CTMI T _{max}
57	57	KR_FIX_CYN_20	Fixing cyanobacteria RESP rate constant
58	58	THETA_KR_FIX_C	Fixing cyanobacteria Temperature correction for RESP rate
59	59	KD_FIX_CYN_20	Fixing cyanobacteria Mortality rate constant
60	60	THETA_KD_FIX_C	Fixing cyanobacteria Temperature correction for Mortality rate
61	61	KHS_DIN_FIX_CYN	Fixing cyanobacteria Half saturation growth for DIN
62	62	KHS_DIP_FIX_CYN	Fixing cyanobacteria Half saturation growth for DIP
63	63	KHS_O2_FIX_CYN	Fixing cyanobacteria Half saturation growth for O ₂
64	64	FRAC_FIX_CYN_EX	Fixing cyanobacteria Fraction of excretion in metabolism rate
65	65	I_S_FIX_CYN	Fixing cyanobacteria Light saturation (Iangleys). Not used, is calculated!
66	66	DO_STR_HYPOX_FIX	Fixing cyanobacteria Dissolved oxygen stress in oxygen units (mortality increase below this value exponentially)
67	67	THETA_HYPOX_FIX_CYN_D	Fixing cyanobacteria Multiplier of the exponent for Dissolved oxygen stress
68	68	EXPON_HYPOX_FIX_CYN_D	Fixing cyanobacteria Exponent constant for Dissolved oxygen stress
69	69	FX_CYN_N_TO_C	Fixing cyanobacteria Nitrogen to Carbon ratio
70	70	FX_CYN_P_TO_C	Fixing cyanobacteria Phosphorus to Carbon ratio
71	71	FIX_CYN_O2_TO_C	Fixing cyanobacteria Oxygen to Carbon ratio for respiration
72	72	FIX_CYN_C_TO_CHL	Fixing cyanobacteria Carbon to Chlorophyll a ratio
73	73	R_FIX	Fixing cyanobacteria Ratio between non-fixing and fixing fractions growth rate
74	74	K_FIX	Fixing cyanobacteria Effectivity parameter of switching to nitrogen fixation

Literature References:

[1] Staal, M., Meysman, F.J.R. & Stal, L.J. (2003)

Temperature excludes N₂-fixing heterocystous cyanobacteria in the tropical oceans. Nature, 425, 504-507.

Relevance: Temperature controls on N-fixation; optimal range 20-30 deg C.

[2] Horne, A.J. & Goldman, C.R. (1994)

Limnology, 2nd ed. McGraw-Hill.

Relevance: N-fixation rates and conditions triggering fixation when DIN is low.

[3] Paerl, H.W. & Otten, T.G. (2013)

Harmful cyanobacterial blooms: causes, consequences, and controls. Microbial Ecology, 65(4), 995-1010.

Relevance: Growth ecology and bloom dynamics of N-fixing cyanobacteria.

[4] Grimaud, G.M., Mairet, F. et al. (2017)

Modeling the temperature effect on the specific growth rate of phytoplankton: a review. Reviews in Environmental Science and Bio/Technology, 16, 625-645.

Relevance: Review of CTMI and other temperature-growth models for phytoplankton.

Other Phytoplankton (Chlorophyta, Cryptophyta, etc.)

Growth and loss parameters for a generic 'other phytoplankton' group representing green algae, cryptophytes, and chrysophytes.
Typical units: 1/day, deg C, mg/L, dimensionless

#	Constant Name	Value	Description
75	75	KG_OPA_OPT_TE	OtherPhyto Growth rate constant
76	76	OPA_OPT_TEMP_	OtherPhyto CTMI T_min
77	77	OPA_OPT_TEMP_	OtherPhyto CTMI T_opt
78	78	EFF_OPA_GROWTH	OtherPhyto Effective growth. (1-EG)*growth - losses for respiration and excretion
79	79	KAPPA_OPA_UNDER_O	(unused)
80	80	KAPPA_OPA_OVER_OP	OtherPhyto CTMI T_max
81	81	KR_OPA_20	OtherPhyto Respiration rate constant
82	82	THETA_KR_OPA	OtherPhyto Temperature correction for respiration rate
83	83	KD_OPA_20	OtherPhyto Mortality rate constant
84	84	THETA_KD_OPA	OtherPhyto Temperature correction for Mortality rate
85	85	KHS_DIN_OPA	OtherPhyto Half saturation growth for DIN
86	86	KHS_DIP_OPA	OtherPhyto Half saturation growth for DIP
87	87	KHS_O2_OPA	OtherPhyto Half saturation growth for O2
88	88	FRAC_OPA_EXC	OtherPhyto Fraction of excretion in metabolism rate
89	89	I_S_OPA	OtherPhyto Light saturation (langleys). Not used, is calculated!
90	90	DO_STR_HYPOX_O	OtherPhyto Dissolved oxygen stress in oxygen units (mortality increase below this value exponentially)
91	91	THETA_HYPOX_OP	OtherPhyto Multiplier of the exponent for Dissolved oxygen stress
92	92	EXPON_HYPOX_OP	OtherPhyto Exponent constant for Dissolved oxygen stress
93	93	OPA_N_TO_C	OtherPhyto Nitrogen to Carbon ratio
94	94	OPA_P_TO_C	OtherPhyto Phosphorus to Carbon ratio
95	95	OPA_O2_TO_C	OtherPhyto Oxygen to Carbon ratio for respiration
96	96	OPA_C_TO_CHL	OtherPhyto Carbon to Chlorophyll a ratio

Literature References:

[1] Reynolds, C.S. & Irish, A.E. (1997)

Modelling phytoplankton dynamics in lakes and reservoirs: the problem of in-situ growth rates. Hydrobiologia, 349, 5-17.
Relevance: In-situ growth rates for green algae, cryptophytes; mu_max 1.5-3.5 /day.

[2] Sommer, U. (1989)

Plankton Ecology: Succession in Plankton Communities. Springer.
Relevance: OtherPhyto T_opt typically 15-22 deg C (cooler than cyanobacteria).

Zooplankton

Growth, grazing preferences, half-saturation constants, respiration, and mortality for the bulk zooplankton compartment (mainly crustacean micro-and mesozooplankton).

Typical units: 1/day, deg C, mg C/L, dimensionless

#	Constant Name	Value	Description
97	97	KG_ZOO_OPT_TE	Zooplankton Growth rate
98	98	ZOO_OPT_TEMP	Zooplankton CTMI T_min
99	99	ZOO_OPT_TEMP	Zooplankton CTMI T_opt
100	100	EFF_ZOO_GROWTH	Zooplankton Effective growth. (1-EG)*growth - losses for respiration and excretion
101	101	KAPPA_ZOO_UNDER_OF	(unused)
102	102	KAPPA_ZOO_OVER_OF	Zooplankton CTMI T_max
103	103	GRAT_ZOO_DIA	Zooplankton Grazing rate (growth rate multiplier) on diatoms
104	104	GRAT_ZOO_CYN	Zooplankton Grazing rate (growth rate multiplier) on Cyanobacteria
105	105	GRAT_ZOO_OP	Zooplankton Grazing rate (growth rate multiplier) on OtherPhyto
106	106	GRAT_ZOO_FIX_CYN	Zooplankton Grazing rate (growth rate multiplier) on fixing Cyanobacteria
107	107	GRAT_ZOO_NOST_VEG_HET	Zooplankton Grazing rate (growth rate multiplier) on Nostocles (veg + het)
108	108	GRAT_ZOO_DET_PART	Zooplankton Grazing rate (growth rate multiplier) on part. ORG_C
109	109	PREF_ZOO_DIA	Zooplankton Preference for Diatoms
110	110	PREF_ZOO_CYN	Zooplankton Preference for Cyanobacteria
111	111	PREF_ZOO_FIX_C	Zooplankton Preference for fixing Cyanobacteria
112	112	PREF_ZOO_NOST_VE	Zooplankton Preference for nostocales (veg + het)
113	113	PREF_ZOO_OP	Zooplankton Preference for OtherPhyto
114	114	PREF_ZOO_DET_PART	Zooplankton Preference for Part. ORG_C
115	115	KHS_DIA_C_ZO	Zooplankton Half saturation growth for diatoms
116	116	KHS_CYN_C_ZO	Zooplankton Half saturation growth for Cyanobacteria
117	117	KHS_FIX_CYN_C_Z	Zooplankton Half saturation growth for fixing Cyanobacteria
118	118	KHS_NOST_VEG_HET	Zooplankton Half saturation growth for Nostocales (veg + het)
119	119	KHS_OPA_C_ZO	Zooplankton Half saturation growth for OtherPhyto
120	120	KHS_DET_PART_ORG	Zooplankton Half saturation growth for part. ORG_C
121	121	FOOD_MIN_ZOO	Zooplankton Minimum food conc. for feeding
122	122	KE_ZOO	not used Zooplankton Excretion rate as growth fraction
123	123	FRAC_ZOO_EX_O	not used Zooplankton Excretion rate organic fraction
124	124	KR_ZOO_20	Zooplankton Respiration rate
125	125	THETA_KR_ZOO	Zooplankton Respiration rate Temperature correction
126	126	KD_ZOO_20	Zooplankton Mortality rate
127	127	THETA_KD_ZOO	Zooplankton Mortality rate Temperature correction
128	128	DO_STR_HYPOX_ZO	Zooplankton Dissolved oxygen stress in oxygen units (mortality increase below this value exponential)
129	129	THETA_HYPOX_ZOO_D	Zooplankton Multiplier of the exponent for Dissolved oxygen stress
130	130	EXPON_HYPOX_ZO	Zooplankton Exponent constant for Dissolved oxygen stress
131	131	ZOO_N_TO_C	Zooplankton Nitrogen to Carbon ratio
132	132	ZOO_P_TO_C	Zooplankton Phosphorus to Carbon ratio
133	133	ZOO_O2_TO_C	Zooplankton Oxygen to Carbon ratio for respiration

Literature References:

[1] Jorgensen, S.E. (1995)

The growth rate of zooplankton at the edge of chaos: ecological models. J. Theoretical Biology, 175(1), 13-21.

Relevance: Zooplankton growth rates 0.2-0.8 /day; model uses 0.45 /day.

[2] Hansen, P.J., Bjornsen, P.K. & Hansen, B.W. (1997)

Zooplankton grazing and growth: scaling within the 2-2000 um body size range. Limnology and Oceanography, 42, 687-704.

Relevance: Grazing rates scaling with body size; half-saturation for food 0.02-0.5 mg C/L.

[3] Jorgensen, S.E. (1976)

A eutrophication model for a lake. Ecological Modelling, 2(2), 147-165.

Relevance: Threshold food concentration for zooplankton grazing (FOOD_MIN ~ 0.02 mg C/L). Zooplankton N:C ~ 0.17-0.25, P:C ~ 0.015-0.03.

[4] Sommer, U. & Sommer, F. (2006)

Cladocerans versus copepods: the cause of contrasting top-down controls on freshwater and marine phytoplankton. Oecologia, 147, 183-194.

Relevance: Selective feeding preferences in freshwater zooplankton communities.

Particulate Organic Matter (POM) Dissolution

First-order dissolution rates for particulate organic carbon (POC), nitrogen (PON), phosphorus (POP), and biogenic silica (BSi). Phytoplankton-enhancement factors simulate bacterial colonisation.

Typical units: 1/day, dimensionless

#	Constant Name	Value	Description
134	KDISS_DET_PART_OR		POC Dissolution rate not dependent on phytoplankton
135	THETA_KDISS_DET_PAR		POC Carbon Dissolution rate Temperature correction
136	FAC_PHYT_DET_PART_ORG_C	!0.001	POC Carbon Phytoplankton linear factor for dissolution rate
137	KDISS_DET_PART_OR		PON Dissolution rate not dependent on phytoplankton
138	THETA_KDISS_DET_PAR		PON Dissolution rate Temperature correction
139	KHS DISS_N		PON dissolution reverse half saturation for DIN
140	FAC_PHYT_DET_PART_ORG_N	!0.007	PON Phytoplankton linear factor for dissolution rate
141	KDISS_DET_PART_OR		POP Phosphorus Dissolution rate not dependent on phytoplankton
142	THETA_KDISS_DET_PAR		POP Dissolution rate Temperature correction
143	KHS DISS_P		POP Dissolution reverse half saturation for DIP
144	FAC_PHYT_DET_PART_ORG_P	!0.005	POP Phytoplankton linear factor for dissolution rate
145	KDISS_PART_Si		Particulate Silica Dissolution rate
146	THETA_KDISS_PART_Si		Particulate Silica Dissolution rate Temperature correction

Literature References:

[1] Wetzel, R.G. (2001)

Limnology: Lake and River Ecosystems, 3rd ed. Academic Press.

Relevance: POM dissolution rates: POC 0.01-0.1 /day; PON 0.05-0.5 /day; POP 0.1-1.0 /day.

[2] Bowie, G.L. et al. (1985)

As above -- EPA compilation.

Relevance: Temperature correction theta 1.04-1.08 for organic matter dissolution.

[3] Kamatani, A. (1982)

Dissolution rates of silica from diatoms decomposing at various temperatures. Marine Biology, 68, 91-96.

Relevance: Biogenic silica dissolution rate 0.001-0.01 /day, increasing with temperature.

Dissolved Organic Matter (DOM) Mineralisation

Phytoplankton-dependent mineralisation rate factors and reverse half-saturation concentrations for DOC, DON, and DOP.
Typical units: 1/day, mg/L, dimensionless

#	Constant Name	Value	Description
147	147	FAC_PHYT_AMIN_D	0.045 DOC Phytoplankton linear factor for mineralisation rate
148	148	KHS_AMIN_N	DON mineralisation reverse half saturation for DIN
149	149	FAC_PHYT_AMIN_D	DON Phytoplankton linear factor for mineralisation rate
150	150	KHS_AMIN_P	DOP reverse half saturation for DIP
151	151	FAC_PHYT_AMIN_D	DOP Phytoplankton linear factor for mineralisation rate

Literature References:

[1] Chapra, S.C. (1997)

Surface Water-Quality Modeling. McGraw-Hill.
Relevance: DOM mineralisation rates and phytoplankton enhancement factors.

[2] Bowie, G.L. et al. (1985)

As above.
Relevance: DOC mineralisation rates 0.01-0.2 /day; DON 0.005-0.1 /day.

Nitrification

Ammonia oxidation (nitrification) rate, temperature correction, half-saturation for O₂ and NH₄-N, and pH optimum range.

Typical units: 1/day, mg O₂/L, mg N/L, pH units

#	Constant Name	Value	Description
152	152	K_NITR_20	Amonia nitrification rate
153	153	THETA_K_NITR	Amonia nitrification rate Temperature constant
154	154	KHS_NITR_OXY	Amonia nitrification half saturation for Oxygen
155	155	KHS_NITR_NH4	Amonia nitrification half saturation for Amonia
156	156	PH_NITR_NH4_M	optimum lower range for pH correction factor for nitrification
157	157	PH_NITR_NH4_M	optimum upper range for pH correction factor for nitrification

Literature References:

[1] Chapra, S.C. (1997)

Surface Water-Quality Modeling. McGraw-Hill.

Relevance: Nitrification rate 0.1-1.0 /day at 20 deg C; theta = 1.04-1.08.

[2] Bowie, G.L. et al. (1985)

As above.

Relevance: KHS_O2 for nitrification 0.5-4.0 mg/L; KHS_NH4 0.01-0.10 mg N/L.

[3] Painter, H.A. (1970)

A review of literature on inorganic nitrogen metabolism in microorganisms. Water Research, 4(6), 393-450.

Relevance: pH optimum range for nitrification 6.5-8.5; strong inhibition below pH 6.0.

[4] Cerco, C.F. (2000)

Phytoplankton kinetics in the Chesapeake Bay eutrophication model. Water Quality and Ecosystems Modeling, 1, 5-49.

Relevance: Nitrification parameters used in major estuary models.

Redox Metal Chemistry (Fe, Mn)

Oxidation/reduction rate constants for Fe(II)/Fe(III) and Mn(II)/Mn(IV), plus reversed Monod half-saturation for DOXY inhibition.
Typical units: 1/day, mg/L

#	Constant Name	Value	Description
158	158	k_OX_FE_II	Oxidation rate for iron 2+
159	159	k_RED_FE_III	reduction rate for iron 3+
160	160	k_OX_MN_II	oxidation rate for manganese 2+
161	161	k_RED_MN_IV	reduction rate for manganese 4+
162	162	KHS_DOXY_FE_III	reversed Monod half saturation of DOXY for iron 3+ reduction
163	163	KHS_DOXY_MN_IV	reversed Monod half saturation of DOXY for manganese 4+ reduction

Literature References:

[1] Stumm, W. & Morgan, J.J. (1996)

Aquatic Chemistry: Chemical Equilibria and Rates in Natural Waters, 3rd ed. Wiley.

Relevance: Fe(II) oxidation rate 10^{-3} to 10^{-1} /day; Mn(II) oxidation 10^{-2} to 10^{-1} /day.

[2] Davison, W. (1993)

Iron and manganese in lakes. Earth-Science Reviews, 34(2), 119-163.

Relevance: Reduction rates of Fe(III) and Mn(IV) under anoxic conditions 0.5-5.0 /day.

DOC Mineralisation -- Multi-Electron-Acceptor

Mineralisation rate constants of DOC using different terminal electron acceptors (O₂, NO₃, Mn(IV), Fe(III), SO₄, methanogenesis). Includes temperature corrections, half-saturations, inhibition (thermodynamic sequence), and pH optima.

Typical units: 1/day, mg/L, pH units

#	Constant Name	Value	Description
164	164	K_MIN_DOC_DOXY_20	Mineralization rate constant of DOC at 20 C for dissolved oxygen as final electron acceptor
165	165	K_MIN_DOC_NO3N_20	Mineralization rate constant of DOC at 20 C for nitrate as final electron acceptor
166	166	K_MIN_DOC_MN_IV_20	Mineralization rate constant of DOC at 20 C for Mn IV as final electron acceptor
167	167	K_MIN_DOC_FE_III_20	Mineralization rate constant of DOC at 20 C for Fe III as final electron acceptor
168	168	K_MIN_DOC_S_PLUS_6_20	Mineralization rate constant of DOC at 20 C for sulphate as final electron acceptor
169	169	K_MIN_DOC_DOC_20	Mineralization rate constant of DOC at 20 C for DOC itself as final electron acceptor
170	170	THETA_K_MIN_DOC_DOXY	Temperature correction factor for mineralization process rate constant of DOC for dissolved oxygen as final electron acceptor
171	171	THETA_K_MIN_DOC_NO3N	Temperature correction factor for mineralization process rate constant of DOC for nitrate as final electron acceptor
172	172	THETA_K_MIN_DOC_MN_IV	Temperature correction factor for mineralization process rate constant of DOC for MN IV as final electron acceptor
173	173	THETA_K_MIN_DOC_FE_III	Temperature correction factor for mineralization process rate constant of DOC for FE III as final electron acceptor
174	174	THETA_K_MIN_DOC_S_PLUS_6	Temperature correction factor for mineralization process rate constant of DOC for Suphate as final electron acceptor
175	175	THETA_K_MIN_DOC_DOC	Temperature correction factor for mineralization process rate constant of DOC for DOC
176	176	K_HS_DOC_MIN_DOXY	Monod type half-saturation concentration of DOC for DOC mineralization for dissolved oxygen as the final electron acceptor
177	177	K_HS_DOC_MIN_NO3N	Monod type half-saturation concentration of DOC for DOC mineralization for nitrate as the final electron acceptor
178	178	K_HS_DOC_MIN_MN_IV	Monod type half-saturation concentration of DOC for DOC mineralization for Mn IV as the final electron acceptor
179	179	K_HS_DOC_MIN_FE_III	Monod type half-saturation concentration of DOC for DOC mineralization for Fe III as the final electron acceptor
180	180	K_HS_DOC_MIN_S_PLUS_6	Monod type half-saturation concentration of DOC for DOC mineralization for sulphate as the final electron acceptor
181	181	K_HS_DOC_MIN_DOC	Monod type half-saturation concentration of DOC for DOC mineralization for DOC itself as the final electron acceptor
182	182	K_HS_DOXY_RED	Monod type half-saturation concentration of dissolved oxygen to limit the consumption of dissolved oxygen as the final electron acce
183	183	K_HS_NO3N_RED	Monod type half-saturation concentration of nitrate nitrogen to limit the consumption of nitrate as the final electron acceptor for
184	184	K_HS_MN_IV_RED	Monod type half-saturation concentration of MN_IV to limit the consumption of Mn IV as the final electron acceptor for DOC mineraliz
185	185	K_HS_FE_III_RED	Monod type half-saturation concentration of FE_III to limit the consumption of Fe III as the final electron acceptor for DOC mineral
186	186	K_HS_S_PLUS_6_RED	Monod type half-saturation concentration of sulphate to limit the consumption of sulphate as the final electron acceptor for DOC min
187	187	K_HS_DOXY_RED_I	Reversed Monod type half saturation concentration that simulate the inhibition effect of dissolved oxygen on consumption of NO ₃ , Mn
188	188	K_HS_NO3N_RED_I	Reversed Monod type half saturation concentration that simulate the inhibition effect of nitrate on consumption of Mn IV, Fe III, S

#	Constant Name	Value	Description
189	189	K_HS_MN_IV_RED	Reversed Monod type half saturation concentration that simulate the inhibition effect of Mn IV on consumption of Fe III, S VI and DO
190	190	K_HS_FE_III_RED_I	Reversed Monod type half saturation concentration that simulate the inhibition effect of Fe III on consumption of S VI and DOC itself
191	191	K_HS_S_PLUS_6_RED	Reversed Monod type half saturation concentration that simulate the inhibition effect of S VI on consumption of DOC itself for DOC m
192	192	PH_MIN_DOC_MIN_DOXY	Min. pH for the optimum pH range for DOC mineralization with DOXY as final electron acceptor
193	193	PH_MIN_DOC_MIN_NO3N	Min. pH for the optimum pH range for DOC mineralization with NO3N as final electron acceptor
194	194	PH_MIN_DOC_MIN_MN_IV	Min. pH for the optimum pH range for DOC mineralization with MN_IV as final electron acceptor
195	195	PH_MIN_DOC_MIN_FE_III	Min. pH for the optimum pH range for DOC mineralization with FE_III as final electron acceptor
196	196	PH_MIN_DOC_MIN_S	Min. pH for the optimum pH range for DOC mineralization with S_PLUS_6 as final electron acceptor
197	197	PH_MIN_DOC_MIN_DOC	Min. pH for the optimum pH range for DOC mineralization with DOC as final electron acceptor
198	198	PH_MAX_DOC_MIN	Max. pH for the optimum pH range for DOC mineralization with dissolved oxygen as final electron acceptor
199	199	PH_MAX_DOC_MIN_NO3N	Max. pH for the optimum pH range for DOC mineralization with NO3N as final electron acceptor
200	200	PH_MAX_DOC_MIN_MN_IV	Max. pH for the optimum pH range for DOC mineralization with MN_IV as final electron acceptor
201	201	PH_MAX_DOC_MIN_FE_III	Max. pH for the optimum pH range for DOC mineralization with FE_III as final electron acceptor
202	202	PH_MAX_DOC_MIN_S	Max. pH for the optimum pH range for DOC mineralization with S_PLUS_6 as final electron acceptor
203	203	PH_MAX_DOC_MIN_DOC	Max. pH for the optimum pH range for DOC mineralization with DOC as final electron acceptor

Literature References:

[1] Van Cappellen, P. & Wang, Y. (1996)

Cycling of iron and manganese in surface sediments: a general theory for the coupled transport and reaction of carbon, oxygen, nitrogen, sulfur, iron, and manganese. American Journal of Science, 296, 197-243.

Relevance: Thermodynamic sequence of electron acceptors: O₂ > NO₃ > Mn(IV) > Fe(III) > SO₄ > CH₄. Inhibition half-saturation constants 0.01-1.0 mg/L.

[2] Berner, R.A. (1980)

Early Diagenesis: A Theoretical Approach. Princeton Univ. Press.

Relevance: Foundation for multi-electron-acceptor organic matter degradation kinetics.

[3] Soetaert, K., Herman, P.M.J. & Middelburg, J.J. (1996)

A model of early diagenetic processes from the shelf to abyssal depths. Geochimica et Cosmochimica Acta, 60(6), 1019-1040.

Relevance: DOC mineralisation rates 0.01-0.10 /day; theta 1.02-1.08.

[4] Boudreau, B.P. (1997)

Diagenetic Models and Their Implementation. Springer.

Relevance: pH correction factors for organic matter mineralisation (pH 6-9 optimum).

DON Mineralisation -- Multi-Electron-Acceptor

Analogous to DOC mineralisation but for dissolved organic nitrogen.

Typical units: 1/day, mg/L, pH units

#	Constant Name	Value	Description
204	204	K_MIN_DON_DOXY ₂₀	Mineralization rate constant of DON at 20 C for dissolved oxygen as final electron acceptor
205	205	K_MIN_DON_NO3N ₂₀	Mineralization rate constant of DON at 20 C for NO3_N as final electron acceptor
206	206	K_MIN_DON_MN_IV ₂₀	Mineralization rate constant of DON at 20 C for Mn IV as final electron acceptor
207	207	K_MIN_DON_FE_III ₂₀	Mineralization rate constant of DON at 20 C for FE III as final electron acceptor
208	208	K_MIN_DON_S_PLUS _{6_20}	Mineralization rate constant of DON at 20 C for S_PLUS_6 as final electron acceptor
209	209	K_MIN_DON_DOC ₂₀	Mineralization rate constant of DON at 20 C for DOC as final electron acceptor
210	210	THETA_K_MIN_DON_DOXY	Temperature correction factor for mineralization process rate constant of DON for dissolved oxygen as final electron acceptor
211	211	THETA_K_MIN_DON_NO3N	Temperature correction factor for mineralization process rate constant of DON for nitrate as final electron acceptor
212	212	THETA_K_MIN_DON_MN_IV	Temperature correction factor for mineralization process rate constant of DON for MN_IV as final electron acceptor
213	213	THETA_K_MIN_DON_FE_III	Temperature correction factor for mineralization process rate constant of DON for FE_III as final electron acceptor
214	214	THETA_K_MIN_DON_S_PLUS_6	Temperature correction factor for mineralization process rate constant of DON for sulphate as final electron acceptor
215	215	THETA_K_MIN_DON_DOC	Temperature correction factor for mineralization process rate constant of DON for DOC
216	216	K_HS_DON_MIN_DON_DOXY	Monod type half-saturation concentration of DON for DON mineralization for dissolved oxygen as the final electron acceptor
217	217	K_HS_DON_MIN_DON_NO3N	Monod type half-saturation concentration of DON for DON mineralization for nitrate nitrogen as the final electron acceptor
218	218	K_HS_DON_MIN_DON_MN_IV	Monod type half-saturation concentration of DON for DON mineralization for MN_IV as the final electron acceptor
219	219	K_HS_DON_MIN_DON_FE_III	Monod type half-saturation concentration of DON for DON mineralization for FE_III as the final electron acceptor
220	220	K_HS_DON_MIN_DON_S_PLUS_6	Monod type half-saturation concentration of DON for DON mineralization for sulphate sulphur as the final electron acceptor
221	221	K_HS_DON_MIN_DON_DOC	Monod type half-saturation concentration of DON for DON mineralization for DOC as the final electron acceptor
222	222	PH_MIN_DON_MIN_DON_DOXY	Min. pH for the optimum pH range for DON mineralization with DOXY as final electron acceptor
223	223	PH_MIN_DON_MIN_DON_NO3N	Min. pH for the optimum pH range for DON mineralization with NO3_N as final electron acceptor
224	224	PH_MIN_DON_MIN_DON_MN_IV	Min. pH for the optimum pH range for DON mineralization with MN IV as final electron acceptor
225	225	PH_MIN_DON_MIN_DON_FE_III	Min. pH for the optimum pH range for DON mineralization with FE III as final electron acceptor
226	226	PH_MIN_DON_MIN_DON_S_PLUS_6	Min. pH for the optimum pH range for DON mineralization with S_PLUS_6 as final electron acceptor
227	227	PH_MIN_DON_MIN_DON_DOC	Min. pH for the optimum pH range for DON mineralization with DOC as final electron acceptor
228	228	PH_MAX_DON_MIN_DON_DOXY	Max. pH for the optimum pH range for DON mineralization with dissolved oxygen as final electron acceptor
229	229	PH_MAX_DON_MIN_DON_NO3N	Max. pH for the optimum pH range for DON mineralization with NO3_N as final electron acceptor
230	230	PH_MAX_DON_MIN_DON_MN_IV	Max. pH for the optimum pH range for DON mineralization with Mn IV as final electron acceptor

#	Constant Name	Value	Description
231	231	PH_MAX_DON_MIN_FE_III	Max. pH for the optimum pH range for DON mineralization with FE_III as final electron acceptor
232	232	PH_MAX_DON_MIN_S_PLUS_6	Max. pH for the optimum pH range for DON mineralization with S_PLUS_6 as final electron acceptor
233	233	PH_MAX_DON_MIN_DOC	Max. pH for the optimum pH range for DON mineralization with DOC as final electron acceptor

Literature References:**[1] Soetaert, K. et al. (1996)**

As above.

Relevance: DON mineralisation rates generally lower than DOC; 0.001-0.1 /day.

[2] Berner, R.A. (1980)

As above.

Relevance: Multi-electron-acceptor framework applies equally to DON.

DOP Mineralisation -- Multi-Electron-Acceptor

Analogous to DOC mineralisation but for dissolved organic phosphorus.

Typical units: 1/day, mg/L, pH units

#	Constant Name	Value	Description
234	234	K_MIN_DOP_DOXY ₂₀	Mineralization rate constant of DOP at 20 C for dissolved oxygen as final electron acceptor
235	235	K_MIN_DOP_NO3N ₂₀	Mineralization rate constant of DOP at 20 C for NO3_N as final electron acceptor
236	236	K_MIN_DOP_MN_IV ₂₀	Mineralization rate constant of DOP at 20 C for Mn IV as final electron acceptor
237	237	K_MIN_DOP_FE_III ₂₀	Mineralization rate constant of DOP at 20 C for FE_III as final electron acceptor
238	238	K_MIN_DOP_S_PLUS ₂₀	Mineralization rate constant of DOP at 20 C for S_PLUS_6 as final electron acceptor
239	239	K_MIN_DOP_DOC ₂₀	Mineralization rate constant of DOP at 20 C for DOC as final electron acceptor
240	240	THETA_K_MIN_DOP_DOXY ₂₀	Temperature correction factor for mineralization process rate constant of DOP for dissolved oxygen as final electron acceptor
241	241	THETA_K_MIN_DOP_NO3N ₂₀	Temperature correction factor for mineralization process rate constant of DOP for nitrate as final electron acceptor
242	242	THETA_K_MIN_DOP_MN_IV ₂₀	Temperature correction factor for mineralization process rate constant of DOP for Mn IV as final electron acceptor
243	243	THETA_K_MIN_DOP_FE_III ₂₀	Temperature correction factor for mineralization process rate constant of DOP for FE_III as final electron acceptor
244	244	THETA_K_MIN_DOP_S_PLUS ₂₀	Temperature correction factor for mineralization process rate constant of DOP for sulphate as final electron acceptor
245	245	THETA_K_MIN_DOP_DOC ₂₀	Temperature correction factor for mineralization process rate constant of DOP for DOC
246	246	K_HS_DOP_MIN_D	Monod type half-saturation concentration of DOP for DOP mineralization for dissolved oxygen as the final electron acceptor
247	247	K_HS_DOP_MIN_N	Monod type half-saturation concentration of DOP for DOP mineralization for nitrate nitrogen as the final electron acceptor
248	248	K_HS_DOP_MIN_M	Monod type half-saturation concentration of DOP for DOP mineralization for MN_IV as the final electron acceptor
249	249	K_HS_DOP_MIN_F	Monod type half-saturation concentration of DOP for DOP mineralization for FE_III as the final electron acceptor
250	250	K_HS_DOP_MIN_S_P	Monod type half-saturation concentration of DOP for DOP mineralization for sulphate sulphur as the final electron acceptor
251	251	K_HS_DOP_MIN_D	Monod type half-saturation concentration of DOP for DOP mineralization for DOC as the final electron acceptor
252	252	PH_MIN_DOP_MIN_D	Min. pH for the optimum pH range for DOP mineralization with dissolved oxygen as final electron acceptor
253	253	PH_MIN_DOP_MIN_N	Min. pH for the optimum pH range for DOP mineralization with NO3_N as final electron acceptor
254	254	PH_MIN_DOP_MIN_M	Min. pH for the optimum pH range for DOP mineralization with Mn IV as final electron acceptor
255	255	PH_MIN_DOP_MIN_F	Min. pH for the optimum pH range for DOP mineralization with FE III as final electron acceptor
256	256	PH_MIN_DOP_MIN_S	Min. pH for the optimum pH range for DOP mineralization with S_PLUS_6 as final electron acceptor
257	257	PH_MIN_DOP_MIN_D	Min. pH for the optimum pH range for DOP mineralization with DOC as final electron acceptor
258	258	PH_MAX_DOP_MIN_D	Max. pH for the optimum pH range for DOP mineralization with dissolved oxygen as final electron acceptor
259	259	PH_MAX_DOP_MIN_N	Max. pH for the optimum pH range for DOP mineralization with NO3_N as final electron acceptor
260	260	PH_MAX_DOP_MIN_M	Max. pH for the optimum pH range for DOP mineralization with Mn IV as final electron acceptor

#	Constant Name		Value	Description
261	261	PH_MAX_DOP_MIN_Fe_III		Max. pH for the optimum pH range for DOP mineralization with Fe III as final electron acceptor
262	262	PH_MAX_DOP_MIN_S		Max. pH for the optimum pH range for DOP mineralization with S_PLUS_6 as final electron acceptor
263	263	PH_MAX_DOP_MIN_DOC		Max. pH for the optimum pH range for DOP mineralization with DOC as final electron acceptor

Literature References:

[1] Soetaert, K. et al. (1996)

As above.
Relevance: DOP mineralisation typically faster than DON due to enzymatic hydrolysis.

[2] Berner, R.A. (1980)

As above.
Relevance: Phosphorus regeneration coupled to redox chemistry.

Methane & Hydrogen Sulphide

Oxidation rates for CH₄ and H₂S with temperature corrections and half-saturation for dissolved oxygen.

Typical units: 1/day, mg/L

#	Constant Name	Value	Description
264	264	k_OX_CH4	Methane oxidation rate constant at 20 C
265	265	THETA_k_OX_CH4	Arrhenius type temperature correction for methane oxidation rate constant
266	266	k_HS_OX_CH4_DOxy	Monod type Half saturation of methane oxidation for dissolved oxygen
267	267	k_OX_H2S	Hydrogen sulphide oxidation rate constant at 20 C
268	268	THETA_k_OX_H2S	Arrhenius type temperature correction for sulphide oxidation rate constant
269	269	k_HS_OX_H2S_DOxy	Monod type Half saturation of hydrogen sulphide oxidation for dissolved oxygen

Literature References:

[1] Bastviken, D. et al. (2004)

Methane emissions from lakes: dependence of lake characteristics, two regional assessments, and a global estimate. Global Biogeochemical Cycles, 18(4), GB4009.

Relevance: Methane oxidation rates 0.1-10 /day; dependent on O₂ availability.

[2] Jorgensen, B.B. (1982)

Mineralization of organic matter in the sea bed -- the role of sulphate reduction. Nature, 296, 643-645.

Relevance: H₂S oxidation rates in aquatic systems; 0.1-5.0 /day.

Fe Dissolution & Fractionation

Dissolution rate constants for particulate Fe(II) and Fe(III) phases and initial dissolved fractions.
Typical units: 1/day, dimensionless

#	Constant Name	Value	Description
270	270	k DISS_Fe_II_20	Dissolution rate constant for Fe II at 20 C
271	271	THETA_k DISS_Fe_II_20	Dissolution rate constant for Fe II
272	272	INIT_MULT_Fe_II_20	Initial fraction of dissolved Fe II
273	273	k DISS_Fe_III_20	Dissolution rate constant for Fe III at 20 C
274	274	THETA_k DISS_Fe_III_20	Dissolution rate constant for Fe III
275	275	INIT_MULT_Fe_III_20	Initial fraction of dissolved Fe III

Literature References:

[1] Stumm, W. & Morgan, J.J. (1996)
As above.
Relevance: Dissolved vs. particulate Fe fractionation; dissolution rates 0.01-1.0 /day.

Nostocales (Heterocystous Cyanobacteria)

Growth, loss, and akinete (resting stage) formation/germination parameters for Nostocales (*Dolichospermum*, *Aphanizomenon flos-aquae*). Includes density-dependent mortality and seasonal triggers.

Typical units: 1/day, deg C, mg/L, day-of-year

#	Constant Name	Value	Description
276	276	KG_NOST_VEG_HET_O	Nostocales (veg + het) Growth rate constant
277	277	NOST_VEG_HET_OPT_T	Nostocales (veg + het) CTMI T_min
278	278	NOST_VEG_HET_OPT_T	Nostocales (veg + het) CTMI T_opt
279	279	EFF_NOST_VEG_HET_GROWTH	Nostocales (veg + het) Effective growth. (1-EG)*growth - losses for RESP and excretion
280	280	KAPPA_NOST_VEG_HET_UND	(unused)
281	281	KAPPA_NOST_VEG_HET_OVE	Nostocales (veg + het) CTMI T_max
282	282	KR_NOST_VEG_HE	Nostocales (veg + het) RESP rate constant
283	283	THETA_KR_NOST_VE	Nostocales (veg + het) Temperature correction for RESP rate
284	284	KD_NOST_VEG_HE	Nostocales (veg + het) Mortality rate constant
285	285	THETA_KD_NOST_VE	Nostocales (veg + het) Temperature correction for Mortality rate
286	286	M_DENS_VEG_HET	Nostocales (veg + het) mortality rate constant because of too dense population
287	287	KHS_DP_NOST_VEG	Nostocales (veg + het) Half saturation growth for DP
288	288	KHS_O2_NOST_VEG	Nostocales (veg + het) Half saturation growth for O2
289	289	FRAC_NOST_VEG_HE	Nostocales (veg + het) Fraction of excretion in metabolism rate
290	290	I_S_NOST_VEG_HET	Nostocales (veg + het) Light saturation (langleys). Not used, is calculated!
291	291	DO_STR_HYPOX_NOST_V	Nostocales (veg + het) Dissolved oxygen stress in oxygen units (mortality increase below this value exponentially)
292	292	THETA_HYPOX_NOST_VEG_HET_D	Nostocales (veg + het) Multiplier of the exponent for Dissolved oxygen stress
293	293	EXPON_HYPOX_NOST_VEG_HET_D	Nostocales (veg + het) Exponent constant for Dissolved oxygen stress
294	294	NOST_N_TO_C	Nostocales (veg + het) Nitrogen to Carbon ratio
295	295	NOST_P_TO_C	Nostocales (veg + het) Phosphorus to Carbon ratio
296	296	NOST_O2_TO_C	Nostocales (veg + het) Oxygen to Carbon ratio for respiration
297	297	NOST_C_TO_CHL	Nostocales (veg + het) Carbon to Chlorophyll a ratio
298	298	P_GERM_AKI	Germination rate constant for akinets, 1/day
299	299	N_GERM_AKI	DIN concentration to trigger germination (below that germination starts)
300	300	P_FORM_AKI	Formation rate constant for akinets
301	301	DAY_FORM_AKI	Day of the year from which formation of akinets is allowed
302	302	T_FORM_AKI	Temperature to trigger formation (below that formation starts)
303	303	K_LOSS_AKI	Loss rate constant for akinets
304	304	K_MORT_AKI_20	Mortality rate constant of akinets at 20 degrees celsius
305	305	THETA_K_MORT_AKI	Arrhenius temperature correction factor for akinete mortality rate constant
306	306	T_GERM_AKI	Temperature to trigger germination when DIN is also low

Literature References:

[1] Paerl, H.W. & Otten, T.G. (2013)

As above.

Relevance: Growth ecology of Nostocales; μ_{max} 0.5-2.0 /day.

[2] Hense, I. & Beckmann, A. (2006)

Towards a model of cyanobacteria life cycle -- effects of growing and resting stages on bloom formation of N₂-fixing species. *Ecological Modelling*, 195, 205-218.

Relevance: Akinete germination rates 0.1-0.5 /day; formation triggered by T < 16 deg C and declining DIN. Model uses similar parameterisation.

[3] Sukenik, A., Hadas, O. & Kaplan, A. (2012)

Invasion of Nostocales (cyanobacteria) to subtropical and temperate freshwater lakes -- physiological, regional, and global driving forces. *Frontiers in Microbiology*, 3, 86.

Relevance: Nostocales temperature requirements T_{opt} 24-28 deg C; T_{min} 15-20 deg C.

[4] Kovacs, A.W., Presing, M. & Voros, L. (2016)

Thermal-dependent growth characteristics for *Cylindrospermopsis raciborskii*. *Aquatic Ecology*, 50, 97-108.

Relevance: CTMI cardinal temperatures for Nostocales species.

Dissolution Saturation & DOM Availability

Half-saturation constants for POM dissolution saturation, available fractions of DON/DOP for Nostocales, and phytoplankton-dependent mineralisation caps.

Typical units: mg/L, dimensionless

#	Constant Name	Value	Description
307	307	KHS_POC DISS_S	
308	308	KHS_PON DISS_S	
309	309	KHS_POP DISS_S	
310	310	frac_avail_DON	
311	311	frac_avail_DOP	
312	312	frac_avail_DON_NO	
313	313	KHS_DN_NOST_VEG_RET	half saturation constant for limitation of nost. growth on nitrogen
314	314	FRAC_FIX_N_FOR_GR_VEG_RET	fraction of fixed nitrogen used for nostacles for growth. 1-frac is released to environment
315	315	FRAC_NOST_GROWTH	fraction of maximum growth rate supported by fixed nitrogen 1-frac - supported by N forms in WC
316	316	K_MIN_PHYT_AMIN	
317	317	K_MIN_PHYT_AMIN	
318	318	K_MIN_PHYT_AMIN	

Literature References:

[1] Wetzel, R.G. (2001)

As above.

Relevance: Saturation kinetics for POM dissolution.

Photoinhibition (beta)

Photoinhibition parameters for each phytoplankton group. 0 = Steele formulation (default); values > 0 increase photoinhibition strength.

Typical units: dimensionless

#	Constant Name	Value	Description
319	319	BETA_DIA	Diatoms photoinhibition (0=Steele default, >0=stronger)
320	320	BETA_CYN	Non-fixing cyanobacteria photoinhibition
321	321	BETA_FIX_CYN	Fixing cyanobacteria photoinhibition
322	322	BETA_OPA	OtherPhyto photoinhibition
323	323	BETA_NOST_VEG	Nostocales photoinhibition

Literature References:

[1] Steele, J.H. (1962)

Environmental control of photosynthesis in the sea. Limnology and Oceanography, 7(2), 137-150.

Relevance: Classic photoinhibition formulation $P/P_{\max} = (I/I_s) \cdot \exp(1 - I/I_s)$. AQUABC default beta = 0 uses this curve; beta > 0 intensifies inhibition.

[2] Platt, T., Gallegos, C.L. & Harrison, W.G. (1980)

Photoinhibition of photosynthesis in natural assemblages of marine phytoplankton. J. Marine Research, 38, 687-701.

Relevance: Extended P-I curve with explicit photoinhibition parameter beta.

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- [1] Bastviken, D. et al. (2004) Methane emissions from lakes: dependence of lake characteristics, two regional assessments, and a global estimate. *Global Biogeochemical Cycles*, 18(4), GB4009.
- [2] Bernard, O. & Remond, B. (2012) Validation of a simple model accounting for light and temperature effect on microalgal growth. *Bioresource Technology*, 123, 520-527.
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- [4] Boudreau, B.P. (1997) *Diagenetic Models and Their Implementation*. Springer.
- [5] Bowie, G.L. et al. (1985) *As above* -- EPA compilation.
- [6] Bowie, G.L., Mills, W.B., Porcella, D.B. et al. (1985) *Rates, Constants, and Kinetics Formulations in Surface Water Quality Modeling* (2nd ed.). EPA/600/3-85/040. U.S. EPA, Athens, GA.
- [7] Brzezinski, M.A. (1985) The Si:C:N ratio of marine diatoms: interspecific variability and the effect of some environmental variables. *J. Phycology*, 21(3), 347-357.
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